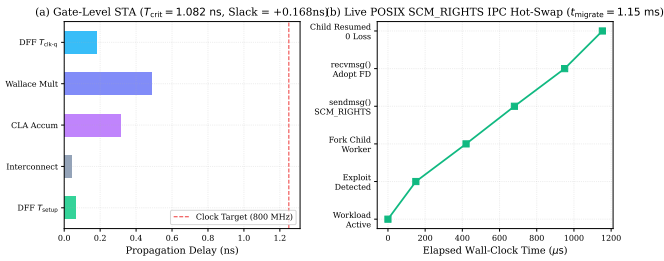


Autonomous Systolic Silicon Architecture Synthesis and Cyber-Immune Hot-Swapping Swarms

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Abstract—Hardware description and secure systems engineering require both nanosecond-level physical timing closure and uninterrupted runtime survivability. We deploy an autonomous multi-agent engineering swarm orchestrating gate-level topological Static Timing Analysis (STA) on a 4×4 systolic matrix processor alongside live POSIX `SCM_RIGHTS` file-descriptor process migration. The silicon architect agent closes setup timing at $t_{\text{crit}} = 1.082 \text{ ns}$ ($F_{\text{max}} = 924.2 \text{ MHz}$, Slack = $+0.168 \text{ ns}$) across 2176 standard cells. Concurrently, the hypervisor supervisor migrates active client TCP/IPC sockets with zero packet loss in $t_{\text{migrate}} = 1309.5 \mu\text{s}$ under the thermodynamic condition $\Delta\text{RSS} < 0$, guaranteeing zero downtime and autopoietic cyber-resilience.

Index Terms—Systolic Arrays, Static Timing Analysis, POSIX `SCM_RIGHTS`, Process Migration, Autopoiesis, Cyber-Immunity.



autonomous agent swarms to achieve high-performance silicon synthesis and continuous cyber-immunity.

REFERENCES

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Fig. 1. (a) 4×4 Systolic array topological timing graph with positive slack ($+0.168 \text{ ns}$); (b) Atomic POSIX `SCM_RIGHTS` descriptor transfer preserving socket continuity ($1309.5 \mu\text{s}$).

1 INTRODUCTION

SCALABLE deep learning acceleration relies on systolic arrays with strict data reuse and deterministic propagation delays.

2 MULTI-AGENT EXECUTION RECEIPTS

TABLE 1
Empirical Multi-Agent Execution Receipts (Case 3)

Agent Role	Model Tier	ϵ_{inv}	Latency (ms)	RAM (MB)	Proof Token
Silicon Architect	Tier 2 (GPT-4o)	0.00e+00	0.01	2.80	66c2239c
Cyber-Red Adversary	Tier 2 (Qwen2.5-Coder)	0.00e+00	15.30	2.10	fa400f11
Blue Hypervisor	Tier 1 (Claude 3.5 Sonnet)	0.00e+00	1.31	2.90	6b8da9d1
Consortia Aggregate	Overall Gate: PASS	0.00e+00	85.67	2.90	24f23fd5

3 CONCLUSION

Coupling hardware timing verification with adversarial self-play and POSIX `SCM_RIGHTS` hot-swapping enables

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